

Linearity of expectation: $E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i]$
 (From Thursday)

we do not need independence

That problem: we have that $X_i \nparallel X_j$, but $E[X_i + X_j] = E[X_i] + E[X_j]$
 is not independent

★ (Today) But $\text{var}(X_i + X_j) \neq \text{var}(X_i) + \text{var}(X_j)$
 If $X_i \perp\!\!\!\perp X_j$ Then $\text{var}(X_i + X_j) = \text{var}(X_i) + \text{var}(X_j)$

	$E[X_i + X_j]$	$\text{var}(X_i + X_j)$
$X_i \perp\!\!\!\perp X_j$	$E[X_i] + E[X_j]$	$\text{var}(X_i) + \text{var}(X_j)$
$X_i \nparallel X_j$	$E[X_i] + E[X_j]$	$\text{var}(X_i) + \text{var}(X_j) + 2\text{cov}(X_i, X_j) \neq \text{var}(X_i) + \text{var}(X_j)$

→ today

Moment generating functions

Recall: The n^{th} moment of a r.v. X is given by $E[X^n]$
 First moment of X is $E[X]$ ($n=1$)
 Second moment of X is $E[X^2]$ ($n=2$)

$\vdots$
 n^{th} moment of X is $E[X^n]$ (general n)
 (LOTUS)

Recall: Law of The unconscious Statistician says that

$$E[g(X)] = \begin{cases} \sum_k g(x_k) p(x_k) & \text{if } X \text{ is discrete} \\ \int g(x) f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Then what is

$$E[X^n] = \begin{cases} \sum_k x_k^n p(x_k) & \text{if } X \text{ is discrete} \\ \int x^n f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

$g(X) = X^n$. Then apply LOTUS

★ one way to compute n^{th} moment of X

Moment generating functions use a way to compute the moments of X without needing to compute (★)

DEF: The moment generating function of a r.v. X is denoted by $M_X(t)$ and is defined as $M_X(t) = E[e^{tx}]$ for all $t \in \mathbb{R}$

$$M_X(t) = \begin{cases} \sum_k e^{tx_k} p(x_k) & \text{if } X \text{ is discrete} \\ \int e^{tx} f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Q: How do we use $M_X(t)$ to find the moments of X ?

① $M_X(t) = \mathbb{E}[e^{tx}]$ Recall: Taylor series of e^y around $y=0$ is given by

$$e^y = 1 + y + \frac{1}{2}y^2 + \frac{1}{6}y^3 + \dots$$

Taylor series/
Taylor expansion
are important
for 5203, 5204, ...

infinite sum

Exercise: show this

② $M_X(t) = \mathbb{E}[1 + tx + \frac{1}{2}(tx)^2 + \dots]$

③ $\frac{d}{dt} M_X(t) = \frac{d}{dt} \mathbb{E}[1 + tx + \frac{1}{2}(tx)^2 + \dots]$

we assume we can do this $\rightarrow$

$$= \mathbb{E}\left[\frac{d}{dt} (1 + tx + \frac{1}{2}(tx)^2 + \dots)\right]$$
$$= \mathbb{E}[X + (tx) + \dots]$$

④ $\frac{d}{dt} M_X(t) \Big|_{t=0} = \mathbb{E}[X]$ $\rightarrow$ all these remaining terms are some multiple of t

Exercise: show that $\frac{d^2}{dt^2} M_X(t) \Big|_{t=0} = \mathbb{E}[X^2]$

...

In general, $\mathbb{E}[X^n] = \frac{d^n}{dt^n} M_X(t) \Big|_{t=0}$

Another way to see this:

$$\left. \begin{aligned} \frac{d}{dt} M_X(t) &= \frac{d}{dt} \mathbb{E}[e^{tx}] \\ &= \mathbb{E}\left[\frac{d}{dt} e^{tx}\right] \\ &= \mathbb{E}[Xe^{tx}] \end{aligned} \right\} \Rightarrow \frac{d}{dt} M_X(t) \Big|_{t=0} = \mathbb{E}[X]$$

moment generating function

THM: Let X and Y be r.v.s with mgf $M_X(t)$ and $M_Y(t)$ (that are both finite in a neighborhood of 0). If $M_X(t) = M_Y(t) \forall t$ in a neighborhood of 0, then X and Y have the same distribution.

Significance: The mgf can uniquely identify the distribution of a r.v.

Aside: The moments of a r.v. do not uniquely identify its distribution

If $X \stackrel{d}{=} Y \Rightarrow \mathbb{E}[X^n] = \mathbb{E}[Y^n]$ for all n (such that the expectations exist)

But $\mathbb{E}[X^n] = \mathbb{E}[Y^n] \not\Rightarrow X \stackrel{d}{=} Y$

whereas $X \stackrel{d}{=} Y \Leftrightarrow M_X(t) = M_Y(t)$ for t around 0

Example: $X \sim \text{Binomial}(n, p)$. Calculate $M_X(t)$ and use it to calculate $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$

uses Binomial Theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

$$M_X(t) = [pe^t + (1-p)]^n$$

$$\mathbb{E}[X] = M'_X(0) = \frac{d}{dt} M_X(t) \Big|_{t=0}$$

$$\mathbb{E}[X^2] = M''_X(0)$$

$$\begin{aligned} \mathbb{E}[e^{tx}] &= \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} \cdot e^{tx} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\ &= [pe^t + (1-p)]^n \end{aligned}$$

$$M_X'(t) = np e^t [pe^t + (1-p)]^{n-1}$$

$$M_X''(t) = np e^t [pe^t + (1-p)]^{n-1}$$

$$+ np e^t ((n-1)[pe^t + (1-p)]^{n-2} pe^t)$$

} product rule,
chain rule

$$E[X] = M_X'(0) = np$$

$$E[X^2] = M_X''(0) = n(n-1)p^2 + np$$

$$M_X''(0) = np [p + (1-p)] + np(n-1)p = np + n(n-1)p^2$$

Bonus: What is $\text{var}(X)$? $\text{var}(X) = E[X^2] - E[X]^2$
 $= n(n-1)p^2 + np - (np)^2 = \dots = np(1-p)$

* We've been assuming that the mgf exists (it does not always exist)

Proposition: The mgf of the sum of two independent r.v.s is the product of their mgfs. If X and Y are independent, then $M_{X+Y}(t) = M_X(t)M_Y(t)$

Proof: $M_{X+Y}(t) \stackrel{\text{DEF}}{=} E[e^{t(X+Y)}] = E[e^{tX+tY}] = E[e^{tX}e^{tY}]$
 $= E[e^{tX}]E[e^{tY}]$ (Independence)
 $= M_X(t)M_Y(t)$

Proposition: The mgf of a linear combination of independent r.v.s $(\sum_{i=1}^n a_i X_i)$ is given by

$$M_{\sum_{i=1}^n a_i X_i}(t) = \prod_{i=1}^n M_{X_i}(a_i t)$$

Proof: $M_{\sum_{i=1}^n a_i X_i}(t) = E[e^{t \sum_{i=1}^n a_i X_i}] = E[e^{\sum_{i=1}^n t a_i X_i}] = E[\prod_{i=1}^n e^{t a_i X_i}]$
 $= \prod_{i=1}^n E[e^{t a_i X_i}]$ (Independence)
 $= \prod_{i=1}^n M_{X_i}(a_i t)$

One more: $M_{\sum_{i=1}^n a_i X_i + b}(t) = e^{tb} \prod_{i=1}^n M_{X_i}(a_i t)$ (exercise)

Useful polynomial expansions:

Binomial theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

Multinomial theorem: $(x_1 + x_2 + \dots + x_r)^n = \sum_{\substack{n_1+n_2+\dots+n_r=n \\ n_1, n_2, \dots, n_r \geq 0}} \binom{n}{n_1, n_2, \dots, n_r} x_1^{n_1} x_2^{n_2} \dots x_r^{n_r}$

Covariance (and Correlation)

Covariance: some notion of dependence between random variables

e.g., X = height, Y = weight

X = weight of a fluid, Y = volume of that fluid

DEF: The covariance between two r.v.s X and Y is denoted by $\text{Cov}(X, Y)$ and is given by $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$
 $= E[XY] - E[X]E[Y]$ (exercise)

Notes

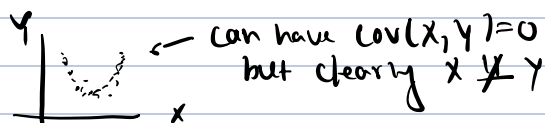
- Suppose $X \perp\!\!\!\perp Y$. Then

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[X]\mathbb{E}[Y] = 0$$

- $\text{Cov}(X, Y) = 0 \not\Rightarrow X \perp\!\!\!\perp Y$

$$X \perp\!\!\!\perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

- "example" suppose



- Counterexample: Define X and Y such that $\mathbb{P}(X=0) = \mathbb{P}(X=1) = \mathbb{P}(X=-1) = \frac{1}{3}$

$$Y = \begin{cases} 0 & \text{if } X \neq 0 \\ 1 & \text{if } X = 0 \end{cases}$$

1. X and Y are not independent *exercise: show rigorously*

2. But $\text{Cov}(X, Y) = 0$

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

$$\mathbb{E}[X] = \frac{1}{3}(0) + \frac{1}{3}(-1) + \frac{1}{3}(1) = 0$$

$$\mathbb{E}[XY] = 0$$

$$XY = \begin{cases} 0 & \text{if } X \neq 0 \\ 0 & \text{if } X = 0 \end{cases}$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{Cov}(X, X) = \text{var}(X)$$

$$\text{Cov}(aX, Y) = \text{Cov}(X, aY) = a \text{Cov}(X, Y) \text{ and } \text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$$

$$\Rightarrow \text{var}(aX) = \text{Cov}(aX, aX) = a \cdot a \text{Cov}(X, X) = a^2 \text{var}(X)$$

$$\text{Cov}\left(\sum_{i=1}^m X_i, \sum_{j=1}^n Y_j\right) = \sum_{i=1}^m \sum_{j=1}^n \text{Cov}(X_i, Y_j)$$

exercise
linearity of covariance

Proof: let's define $\mu_i = \mathbb{E}[X_i] \quad i=1, \dots, m$
 $\nu_j = \mathbb{E}[Y_j] \quad j=1, \dots, n$

$$\text{Cov}\left(\sum_{i=1}^m X_i, \sum_{j=1}^n Y_j\right) = \mathbb{E}\left[\left(\sum_{i=1}^m X_i - \underbrace{\mathbb{E}\left[\sum_{i=1}^m X_i\right]}_{\sum_{i=1}^m \mathbb{E}[X_i]}\right)\left(\sum_{j=1}^n Y_j - \mathbb{E}\left[\sum_{j=1}^n Y_j\right]\right)\right]$$

$$= \sum_{i=1}^m \mu_i$$

$$= \mathbb{E}\left[\left(\sum_{i=1}^m X_i - \sum_{i=1}^m \mu_i\right)\left(\sum_{j=1}^n Y_j - \sum_{j=1}^n \nu_j\right)\right]$$

$$= \mathbb{E}\left[\left(\sum_{i=1}^m (X_i - \mu_i)\right)\left(\sum_{j=1}^n (Y_j - \nu_j)\right)\right]$$

$$= \mathbb{E}\left[\sum_{i=1}^m \sum_{j=1}^n (X_i - \mu_i)(Y_j - \nu_j)\right]$$

$$\sum_{i=1}^m a_i \sum_{j=1}^n b_j \Rightarrow \sum_{i=1}^m \sum_{j=1}^n a_i b_j$$

$$= \sum_{i=1}^m \sum_{j=1}^n \mathbb{E}\left[\underbrace{(X_i - \mu_i)}_{\mathbb{E}[X_i]} \underbrace{(Y_j - \nu_j)}_{\mathbb{E}[Y_j]}\right]$$

may not have $\Leftarrow$

$$= \sum_{i=1}^m \sum_{j=1}^n \text{Cov}(X_i, Y_j)$$

Proposition : $\text{var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j)$

★ covariance can be negative

$$= \sum_{i=1}^n \sum_{j=1}^n \text{cov}(X_i, X_j)$$

In particular, $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y)$

$$\text{var}(X-Y) = \text{var}(X + (-Y))$$

$$= \text{var}(X) + \text{var}(-Y) + 2\text{cov}(X, -Y)$$

$$= \text{var}(X) + (-1)^2 \text{var}(Y) + (-1) 2\text{cov}(X, Y)$$

$$= \text{var}(X) + \text{var}(Y) - 2\text{cov}(X, Y)$$

Also, if $X_i \perp X_j \forall i, j$ (pairwise independent)

Then

$$\text{var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j) \rightarrow 0$$

Correlation coefficient : "scale free version of covariance"

$$\text{cov}(H, W)$$

$$\text{cov}(2H, W) = 2\text{cov}(H, W)$$

DEF : The correlation coefficient between X and Y is denoted by $\rho(X, Y)$ and is given by

↑ "rho"

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)} \sqrt{\text{var}(Y)}}$$

we can show that

$$\rho(X, Y) \in [-1, 1]$$

$$\left(\rho(ax, Y) = \frac{a \text{cov}(X, Y)}{\sqrt{a^2 \text{var}(X)} \sqrt{\text{var}(Y)}} \right)$$